The latent space of equational theories

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Abstract

Building on the collaborative Equational Theories project initiated by Terence Tao fifteen months ago, and combining it with ideas coming from machine learning and finite model theory, we construct a *latent space of equational theories* where each equational theory is located at a specific location, determined by its statistical behavior with respect to a large sample of finite magmas. This experiment enables us to observe for the first time how reasoning flows and produces surprisingly oriented and well-structured chains of logical implications in the latent space of equational theories.

Two things fill the mind with ever new and increasing admiration and awe, the more often and steadily we reflect upon them: the starry heavens above me and the moral law within me.

Immanuel Kant, Critique of Practical Reason.

1 Introduction

This work draws on the recent collaborative project Equational Theories (ET project) initiated by Terence Tao about fifteen months ago (Sept. 2024). In this innovative mathematical experiment, a large community of thirty-three authors exchanging on the Lean Zulip were able to describe *exhaustively* in only a few weeks the full implication preorder $(G, \Rightarrow)$ between 4694 basic equational theories on magmas, see (Bolan et al., 2024) for details.

Taking inspiration from the ET project and using the concept of *Stone pairing* originating from finite model theory (Nešetřil and Ossona de Mendez, 2020), we experiment with the idea that mathematical concepts (in this case, equational theories) can be located as *vertices in a latent space*, and then observed and mapped in the same way as constellations of stars in the sky.

Once the latent space of equational theories has been defined and constructed, we derive an *implication graph* $(G, \rightsquigarrow)$ from the full implication preorder $(G, \Rightarrow)$ made available online by the ET project. This allows us to observe that *reasoning flows in a surprisingly well-organized and oriented way* in the latent space of equational theories.

Despite its simple technical and conceptual design, this empirical study at the crossroad of mathematical logic and machine learning reveals the existence of a rich and well-organised *latent space of equational theories and reasoning flows* at the heart of universal algebra. We believe that the discovery of this emerging piece of mathematical landscape and the exploration of its remarkable geometric and statistical properties provide a nice illustration of the benefits of developing an experimental approach to logic.

Plan of the paper. After recalling in §2 the purpose of the ET project, we explain in §3 how we extract the graph $(G, \rightsquigarrow)$ of atomic steps from the implication preorder $(G, \Rightarrow)$. We then define in §4 the notion of Stone pairing from which we derive our feature space and then our latent space of equational theories in §5 and §6. A number of empirical observations on this latent space and on the flow of implications are made in §6, in §7 and §8. We describe related works and conclude in §9 and §10. We also describe the limitations of our work in §11.

2 The Equational Theories (ET) project

The purpose of the ET project (Bolan et al., 2024) was to describe entirely the implication order $(G, \Rightarrow)$ between the 4694 equational theories defined with *four instances at most* of the binary connective $\diamond$ on a given magma $(A, \diamond)$. This includes fundamental equational theories such as the *associativity law* (numbered as $\mathsf{Eqn}[4512]$ in the project) which involves three variables x, y, z and four instances of the connective:

$$\forall x, y, z \in A, \quad x \diamond (y \diamond z) \;=\; (x \diamond y) \diamond z$$

the *commutativity law* ($\mathsf{Eqn}[43]$) with two variables x, y and two instances of the connective:

$$\forall x, y \in A, \quad x \diamond y \;=\; y \diamond x$$

as well as the *idempotence law* ($\mathsf{Eqn}[3]$) with one variable x and one instance of the connective:

$$\forall x \in A, \quad x = x \diamond x.$$

One methodological benefit of the exhaustive approach advocated by Tao is that other more exotic and largely unexplored equational theories suddenly appear as meaningful. An example is provided by the *Obelix law* ($\mathsf{Eqn}[1491]$) with four instances of the connective:

$$\forall x, y \in A, \quad x = (y \diamond x) \diamond (y \diamond (y \diamond x))$$

and the *Asterix law* ($\mathsf{Eqn}[65]$) with three instances of the connective:

$$\forall x, y \in A, \quad x = (y \diamond (x \diamond (y \diamond x)))$$

Indeed, the Asterix law implies the Obelix law for all *finite magmas* although it is possible to find an *infinite magma* $(A, \diamond)$ which satisfies the Asterix law without satisfying the Obelix law:

$$(A, \diamond) \models \mathsf{Eqn}[65] \quad \text{and} \quad (A, \diamond) \not\models \mathsf{Eqn}[1491].$$

where the symbol $\models$ expresses the fact that the magma $(A, \diamond)$ satisfies a given equational law. This establishes that there is no proof that $\mathsf{Eqn}[65]$ implies $\mathsf{Eqn}[1491]$, see (Bolan et al., 2024) for details.

3 First step: building the graph $(G, \rightsquigarrow)$

3.1 The implication preorder $(G, \Rightarrow)$

We start from the the implication preorder produced by the ET project, which has **4694** elements, one for each equational theory considered in the project. This means that the number of possible implications between equations is

$$4\,694 \times 4\,694 = 22\,033\,636$$

It was established by the members of the ET project that the actual number of implications is

$$\mathbf{8\,178\,279} \approx 0.37 \times 22\,033\,636$$

An implication between equational theories

$$\mathsf{Eqn}[j] \Rightarrow \mathsf{Eqn}[k]$$

is called *reversible* when the converse implication

$$\mathsf{Eqn}[k] \Rightarrow \mathsf{Eqn}[j]$$

holds. We say in that case that the two equational theories are *provably equivalent* and write

$$\mathsf{Eqn}[j] \sim \mathsf{Eqn}[k].$$

This defines an equivalence relation $\sim$ which partitions the set of $4\,694$ vertices into equivalence classes which we call *reversible cliques*. Among the implications of the preorder $(G, \Rightarrow)$, there are

$$\mathbf{5\,702\,669} \approx 0.70 \times 8\,178\,279$$

strict implications and

$$\mathbf{2\,475\,610} \approx 0.30 \times 8\,178\,279$$

reversible implications (including the $4\,694$ self-references).

3.2 The implication graph $(G, \rightsquigarrow)$

Starting from the implication preorder $(G, \Rightarrow)$ provided by the ET project, we construct a directed graph $(G, \rightsquigarrow)$ of *atomic steps*. A strict implication

$$\mathsf{Eqn}[j] \Rightarrow \mathsf{Eqn}[k]$$

is called *atomic* when it cannot be decomposed, in the sense that, for all equational theories $\mathsf{Eqn}[\ell]$,

$$\mathsf{Eqn}[j] \Rightarrow \mathsf{Eqn}[\ell] \quad \text{and} \quad \mathsf{Eqn}[\ell] \Rightarrow \mathsf{Eqn}[k]$$

implies that

$$\mathsf{Eqn}[j] \sim \mathsf{Eqn}[\ell] \quad \text{or} \quad \mathsf{Eqn}[\ell] \sim \mathsf{Eqn}[k].$$

We write in that case:

$$\mathsf{Eqn}[j] \rightsquigarrow \mathsf{Eqn}[k]$$

Note that a strict implication

$$\mathsf{Eqn}[j] \Rightarrow \mathsf{Eqn}[k]$$

holds if and only if there exists a path

$$\mathsf{Eqn}[j] \rightsquigarrow \cdots \rightsquigarrow \mathsf{Eqn}[k]$$

of atomic steps. It is also worth observing that the notion of atomic step is not intrinsic, and depends on the set of equational theories considered in the ET project. The number of atomic steps, or arrows, in the graph $(G, \rightsquigarrow)$ is

$$\mathbf{1\,052\,209} \approx 0.18 \times 5\,702\,669$$

3.3 The implication graph $(G, \rightsquigarrow)$ modulo

One remarkable property of atomic steps is that they are closed under reversible implications, in the sense that every atomic step $\mathsf{Eqn}[j] \rightsquigarrow \mathsf{Eqn}[k]$ and provably equivalent equational theories with the source and target

$$\mathsf{Eqn}[j'] \sim \mathsf{Eqn}[j] \qquad \mathsf{Eqn}[k] \sim \mathsf{Eqn}[k']$$

induces another atomic step $\mathsf{Eqn}[j'] \rightsquigarrow \mathsf{Eqn}[k']$. For that reason, we declare that two atomic steps

$$\mathsf{Eqn}[j] \rightsquigarrow \mathsf{Eqn}[k] \qquad \mathsf{Eqn}[j'] \rightsquigarrow \mathsf{Eqn}[k']$$

are equivalent modulo $\sim$ when $\mathsf{Eqn}[j] \sim \mathsf{Eqn}[j']$ and $\mathsf{Eqn}[k] \sim \mathsf{Eqn}[k']$. We obtain in this way a directed acyclic graph noted $(G, \rightsquigarrow)/\sim$

with **1415** vertices and **4824** arrows

defined as equivalence classes of vertices and arrows of $(G, \rightsquigarrow)$ modulo $\sim$.

4 Stone pairing

We find useful to recall the notion of *Stone pairing* introduced by (Nešetřil and Ossona de Mendez, 2012, 2020) in the field of finite model theory, see also (Gehrke et al., 2022) for a discussion. We will see in §5 and §6 that our definition of the feature space and of the latent space of equational theories

relies on this *probabilistic* variant of validity of a first-order formula φ in a model A. Here, we focus on the *first-order theory of magmas* defined by the signature $\{\diamond : 2\}$ for the binary function $\diamond$. Recall that a magma $(A, \diamond)$ is defined as a set A equipped with a binary operation

$$\diamond \ : \ A \times A \longrightarrow A.$$

A typical first-order predicate φ states the existence of a neutral element:

$$\exists e. \forall x. \quad \big(x \diamond e = x \ \wedge \ e \diamond x = x \big)$$

or the existence of a right residual for any pair of elements x, z of the magma:

$$\forall x. \forall z. \exists y. \quad x \diamond y = z.$$

All equations between magmas can be expressed in that way. Typically, $\mathsf{Eqn}[3721]$ of the equational theory project can be written as the formula

$$\varphi(x, y) \quad \equiv \quad x \diamond y \ = \ (x \diamond y) \diamond (x \diamond y).$$

with two free variables x, y. Now, given a finite magma $(A, \diamond)$ and a first-order formula

$$\varphi(x_1, \ldots, x_n)$$

with n free variables $x_1, \ldots, x_n$, the Stone pairing

$$\langle \, \varphi \, | \, A \, \rangle \ \in \ [0, 1]$$

is the scalar defined as the probability:

$$\frac{\# \left\{ \, (a_1, \ldots, a_n) \in A^n \mid \varphi(a_1, \ldots, a_n) \, \text{holds} \, \right\}}{\# A^n}$$

that a n-tuple of elements

$$(a_1, \ldots, a_n) \in A^n$$

satisfies the n-ary formula φ. Here we write $\# A$ for the cardinal of a finite set A. By way of illustration, observe that

$$A \ \vDash \ \forall \mathbf{x}, \varphi(\mathbf{x}) \quad \Longleftrightarrow \quad \langle \ \varphi(\mathbf{x}) \ | \ A \ \rangle \ = \ 1$$

where we write $A \vDash \psi$ when a first-order formula ψ is satisfied by a given magma A, and use the notation $\mathbf{x} = x_1, \ldots, x_n$ for the sequence of free variables of the formula φ. From this follows in particular that

$$\langle \ x = x \ | \ A \ \rangle \ = \ 1$$

for every finite magma A. Fractional values different from 0 and 1 are the most common in practice. For instance, note that

$$\langle \ x = y \ | \ A \ \rangle \ = \ \frac{1}{\# A}$$

for every finite magma A, since there is a probability of one divided by the cardinal of A that two elements $a_1, a_2 \in A$ happen to be equal.

5 The feature space of equational theories

In this section, we explain how we use the probabilistic concept of Stone pairing in order to construct our latent space of equational theories.

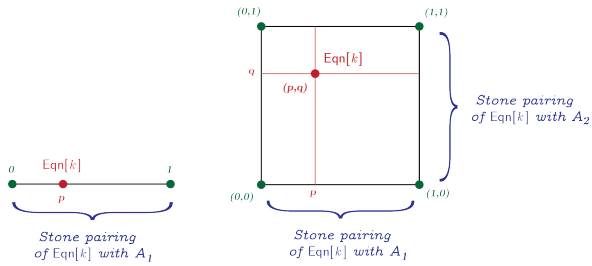

Figure 1: How each equation $\mathsf{Eqn}[k]$ is assigned a position in the hypercube $\mathbb{H}_n$ of dimension $n = \#\mathsf{Mags}$ the number of sampled finite magmas $A_1, \ldots, A_n$.

5.1 The generation process

We proceed in the following way. We start by generating a large number $n = \#\mathsf{Mags}$ of finite magmas $A_1, \ldots, A_n$ of a fixed size N. For simplicity, we define them with the same underlying set $\{0, \ldots, N-1\}$. Each magma A_ℓ for $1 \leq \ell \leq n$ is thus generated as a two-dimensional array of size $N \times N$, where each entry is a randomly chosen value between 0 and $N - 1$. This two-dimensional array describes the multiplication table of the magma $A_\ell = \{0, \ldots, N-1\}$ and thus completely characterizes it. We typically work with a sample of $n = 1000$ magmas of size N between 4 and 16. These numbers $n = 1000$ and $4 \leq N \leq 16$ appear reasonable from a computational point of view and sufficient for our purposes.

5.2 The feature hypercube $\mathbb{H}_n = [0, 1]^n$

We define a matrix of size $\#\mathsf{Eqns} \times \#\mathsf{Mags}$

$$R \quad = \quad \big(p_{k,\ell} \big)_{k \in \mathsf{Eqns}, \ell \in \mathsf{Mags}} \tag{1}$$

where $\#\mathsf{Eqns} = 4964$ is the number of equations and $n = \#\mathsf{Mags} = 1000$ is the number of sampled magmas of size N. Each entry $p_{k,\ell}$ of the matrix R is defined as the probability

$$p_{k,\ell} \quad = \quad \langle \, \mathsf{Eqn}[k] \, | \, A_\ell \, \rangle$$

given by the Stone pairing of $\mathsf{Eqn}[k]$ with the finite magma A_ℓ. Every equation $\mathsf{Eqn}[k]$ induces a line or vector of n probabilities

$$\big(p_{k,1} \ , \ \cdots \ , \ p_{k,n} \big) \ \in \ [0, 1]^n$$

which defines an element of the hypercube $\mathbb{H}_n = [0, 1]^n$ of dimension $n = \#\mathsf{Mags}$, see Fig. 1 for an illustration.

5.3 Stone spectrum

The hypercube $\mathbb{H}_n = [0, 1]^n$ comes equipped with an action